

R PRANAV AJAY



Communication

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Education

● Bachelor of Technology

Amrita Vishwa Vidyapeetham Computer Science and Engineering

2018 - 2022 CGPA: 8.04

● Higher Secondary

Maharishi Intl Residential School Board: CBSE
CGPA: 10 (X)
2015 - 2018 Marks: 455/500 (XII)

Proficiencies

Technical Interests

Linux Automation Game Development
Custom ROM UI&UX Development
Computational Photography
Web & App Development

Tools, Frameworks & Libraries

SimPy	Blender	Project Chrono
ImGui	Carla	Unreal Engine
ROS	gRpc	Godot Engine
Fast API	PyGame	ThreeJS

Programming Languages

C++	GDScript
JavaScript	C
Python	Java

Work Experience

● Caterpillar (Caterpillar India Engineering Solutions Private Limited)

Automation Engineer

- Worked on architecting and implementing pipelines for CI/CD for existing simulation applications and tool chains.
- Worked on deploying a Project Chrono fork in HIL(Hardware in Loop) and SIL (Software in Loop) environments. Aug 2024 - Present
- Worked on profiling the application performance on hardware environment and optimizing the application.

Associate Engineer

- Worked on a Carla fork and was involved on creating scenario generation, implementing newer pipeline for the simulations
- Worked on a Project Chrono fork, implementing the scene graphs, overall simulation pipeline and asset handling pipeline. Aug 2022 - Aug 2024
- Worked on a SimPy fork, implementing scene graph wrappers for non-real time simulations and handling simulation logic.

College Intern

- Worked on an Unreal Engine 4 fork and work on custom scenario generation, asset pipeline and scripting the simulations
- Worked on a pipeline involving Unity and propriety tool chains and focused on creating scene graphs for the simulations Jan 2022 - Jun 2022
- Worked on modelling the 3D assets involved in the simulation pipeline
- Got exposure to the C++ tool chains involved in the pipeline

Projects

Shrula Odyssey

shorturl.at/8lrEp



- A 2D Platformer game with a basic character movement mechanics and side scrolling environment Mar 2025 -Present

Kreative Rum

shorturl.at/8d63u



- A collection of all my projects related to Game Development and Digital Art Nov 2020 -Present

Art Station

shorturl.at/cGQTU



- A collection of my digital art, illustrations and edits Nov 2020 -Present

ToonLyt V2

shorturl.at/dgFKU



- An Android App with Pastiche creation abilities based on Google's project. Dec 2020 - Aug 2021

Content Creator

shorturl.at/cJMZ7



- Created assets and animations for new releases Dec 2020 - Dec 2023
- Administrated the YouTube channel

UI Developer

shorturl.at/ceSZ1



- Designed UI/UX prototypes. Dec 2020 - Dec 2023
- Implement them into the Android Apps.